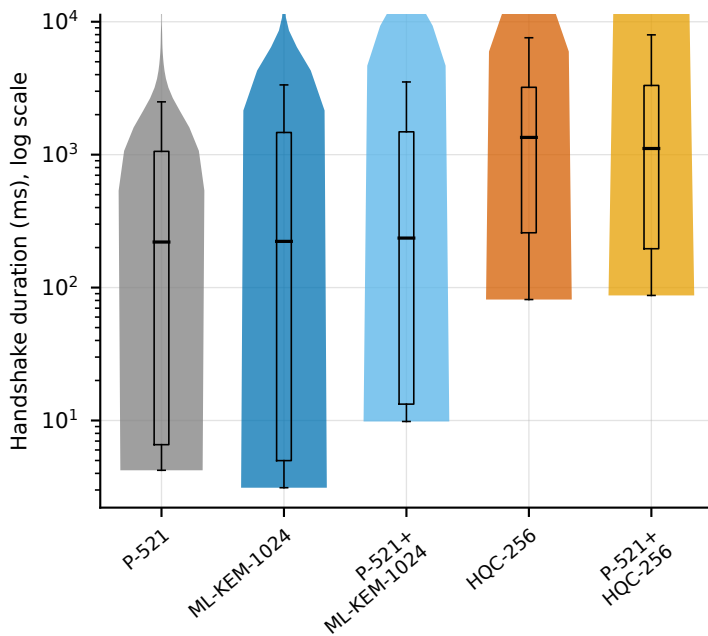
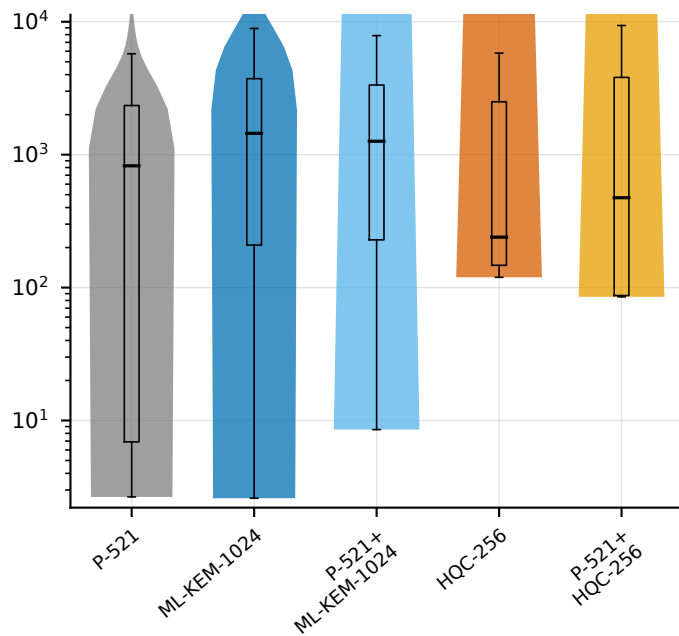


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	1.16k	327.8	52.67k	2.61k	264.8	108.96k
ML-KEM-1024	2.16k	529.5	212.46k	5.36k	322.2	214.43k
P-521+ML-KEM-1024	3.82k	733.9	461.17k	16.88k	1290.6	4.83M
HQC-256	7.18k	451.7	582.55k	11.16k	822.1	1.88M
P-521+HQC-256	8.73k	698.5	1.12M	22.63k	1540.0	7.77M